Straight Line Graphs

Starter

Solution 2

1. Find each intersection.

1(a)

$y = 4x + 7, \quad 3y = 2x - 1$

1(a)

$(-2, -1)$

1(b)

$y = 5x - 1, \quad 3x + 7y = 11$

1(b)

$\left(\frac{9}{19}, \frac{26}{19}\right)$

1(c)

$2x - 5y = -1, \quad 5x - 7y = 14$

1(c)

$(7, 3)$

2. Simplify each surd.

2(a)

$\sqrt{80}$

2(a)

$4\sqrt{5}$

2(b)

$\sqrt{200}$

2(b)

$10\sqrt{2}$

2(c)

$\sqrt{125}$

2(c)

$5\sqrt{5}$

3. Make y the subject.

3(a)

$6x + 3y - 15 = 0$

3(a)

$y = 5 - 2x$

3(b)

$2x - 5y - 9 = 0$

3(b)

$y = \frac{2}{5}x - \frac{9}{5}$

3(c)

$3x - 7y + 12 = 0$

3(c)

$y = \frac{3}{7}x + \frac{12}{7}$

Chapter Overview

Solution 3

Gradients and equations

Use points, gradients and intercepts to describe straight lines.

Parallel and perpendicular lines

Compare gradients and find unknown line equations.

Lengths and areas

Use intersections, Pythagoras and triangle areas.

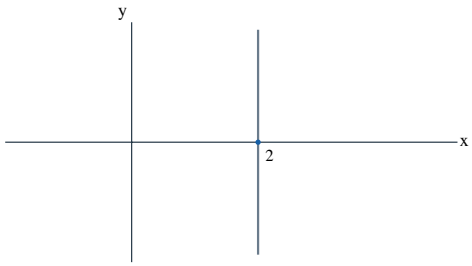
Linear modelling

Interpret gradients and intercepts in practical situations.

Vertical lines

Solution 4

What equation describes this vertical line?



Equation

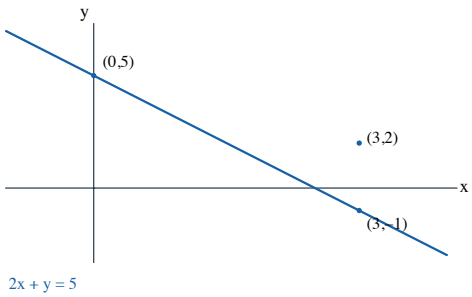
$x = 2$

Every point has x-coordinate 2. The y-coordinate can be any real number.

Lines and Equations of Lines

Solution 5

Which points lie on $2x + y = 5$?



a

$(0, 5)$

a

$2(0) + 5 = 5$

Yes.

b

$(3, 2)$

b

$2(3) + 2 = 8 \neq 5$

No.

c

$(3, -1)$

c

$2(3) - 1 = 5$

Yes.

Examples

Solution 6

1 The point (5,a) lies on $y = 3x + 2$. Find a.

1

$a = 3(5) + 2 = 17$

2 Find where $2x + y = 5$ meets the x-axis.

2

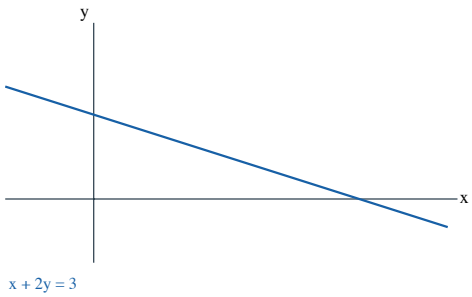
$y = 0 \Rightarrow x = \frac{5}{2}$

$(\frac{5}{2}, 0)$

Test Your Understanding

Solution 7

Find both axis intercepts of $x + 2y = 3$.



Intercepts

$y = 0 \Rightarrow x = 3$

$(3, 0)$

$x = 0 \Rightarrow y = \frac{3}{2}$

$(0, \frac{3}{2})$

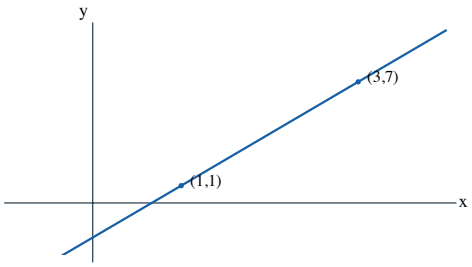
On the x-axis, y=0. On the y-axis, x=0. State an intercept as a coordinate pair.

Recap of gradient

Solution 8

Gradient measures the change in y for each unit change in x.

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$



Example The rise is 6 and the run is 2.

Example

$m = 6/2 = 3$

Subtract the coordinates in the same direction in numerator and denominator. A vertical line has undefined gradient.

Examples

Solution 9

a Find the gradient through $(1,4)$ and $(3,10)$.

a

$$m = \frac{10-4}{3-1} = 3$$

b Find the gradient through $(5,7)$ and $(8,1)$.

b

$$m = \frac{1-7}{8-5} = -2$$

c Find the gradient through $(2,2)$ and $(-1,10)$.

c

$$m = \frac{10-2}{-1-2} = -\frac{8}{3}$$

d Are $(3,4)$, $(5,5)$ and $(11,8)$ collinear?

d

$$\frac{5-4}{5-3} = \frac{8-4}{11-3} = \frac{1}{2}$$

Yes: equal gradients from the same point.

Further Example

Solution 10

Find a The line through $(2,-5)$ and $(4,a)$ has gradient -1.

Find a

$$\frac{a+5}{4-2} = -1 \Rightarrow a+5 = -2 \Rightarrow a = -7$$

y=mx+c

Solution 11

$y = mx + c$

m is the gradient; c is the y-intercept.

Why c? Set x=0.

Why c?

$y = c$

The line meets the y-axis at (0,c).

Why m? Increase x by 1.

Why m?

$[m(x+1)+c] - [mx+c] = m$

The rise is m for a run of 1.

y=mx+c

Solution 12

Example Find the gradient and y-intercept of $4x - 3y + 5 = 0$.

Example

$3y = 4x + 5 \Rightarrow y = \frac{4}{3}x + \frac{5}{3}$

$m = \frac{4}{3}, \quad (0, \frac{5}{3})$

Keep exact fractions rather than rounding.

ax+by+c=0 — Solution 13

Standard form Write $y = \frac{1}{3}x - \frac{2}{3}$ in the form ax+by+c=0 with integer coefficients.

Standard form

$3y = x - 2 \Rightarrow x - 3y - 2 = 0$

Just for your interest... — Solution 14

Why use standard form?

$ax + by + c = 0$

It includes vertical lines such as $x + 4 = 0$, which cannot be written as y=mx+c.

Further mathematics connection

The vector $\binom{a}{b}$ is normal (perpendicular) to the line. For example, $\binom{2}{1}$ is normal to $2x + y = 4$.

$$\begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + c = 0$$

In three dimensions, $ax + by + cz + d = 0$ describes a plane with normal vector (a, b, c), provided that vector is nonzero.

Test Your Understanding

Solution 15

Your turn Write $y = \frac{2}{5}x + \frac{3}{5}$ in standard form.

Your turn

$5y = 2x + 3$

$2x - 5y + 3 = 0$

Exercise 5A – Pure Y1 Page 91

Solution 16

9 Show A(2,3), B(4,4), C(10,7) are collinear.

9

$m_{AB} = \frac{4-3}{4-2} = \frac{1}{2}$

$m_{AC} = \frac{7-3}{10-2} = \frac{1}{2}$

The two lines share A and have equal gradients.

10 Show A(−2a,5a), B(0,4a), C(6a,a) are collinear.

10

$m_{AB} = \frac{-a}{2a} = -\frac{1}{2}, \quad m_{AC} = \frac{-4a}{8a} = -\frac{1}{2}$

For a≠0 the gradients agree. If a=0, all three points coincide at the origin.

Exercise 5B – Pure Y1 Page 92-93 Solution 17

8 A line meets the axes at (5,0) and (0,2). Find its gradient and standard-form equation.

8

$$m = \frac{2-0}{0-5} = -\frac{2}{5}$$

$$2x + 5y - 10 = 0$$

9 Find the gradient and y-intercept of ax+by+c=0, where b≠0.

9

$$y = -\frac{a}{b}x - \frac{c}{b}$$

$$m = -\frac{a}{b}, \quad (0, -\frac{c}{b})$$

10 The line ax−2y+c=0 has gradient 3 and y-intercept 5. Find a and c.

10

$$y = \frac{a}{2}x + \frac{c}{2} \Rightarrow a = 6,\ c = 10$$

Exercise 5B – Pure Y1 Page 92-93 Solution 18

11 A line through (0,6) has gradient -2. Find its intersection with $5x - 8y - 15 = 0$.

11

$$y = -2x + 6$$

$$5x - 8(-2x + 6) - 15 = 0$$

$$21x = 63 \Rightarrow P = (3, 0)$$

12 The lines $y = 3x - 7$ and $ax + 4y - 17 = 0$ meet at (−3,b). Find b and a.

12

$$b = 3(-3) - 7 = -16$$

$$-3a + 4(-16) - 17 = 0$$

$$a = -27$$

Equations using two points/point + gradient

Solution 19

GCSE method Find the line through (3,5) with gradient 2.

GCSE method

$y = 2x + c$

$5 = 2(3) + c \Rightarrow c = -1$

$y = 2x - 1$

A new way...

Solution 20

A fixed point on a line is (x_1, y_1). A variable point is (x,y).

$m = \frac{y - y_1}{x - x_1} \Rightarrow y - y_1 = m(x - x_1)$

Revisit Use the point (3,5) and gradient 2.

Revisit

$y - 5 = 2(x - 3)$

x_1, y_1, m are constants. Keep x and y as variables. This form is especially useful with fractional gradients.

Quickfire Questions

Solution 21

1

$m = 3, \quad P = (1, 2)$

1

$y - 2 = 3(x - 1)$

2

$m = 5, \quad P = (3, 0)$

2

$y = 5(x - 3)$

3

$m = 2, \quad P = (-3, 4)$

3

$y - 4 = 2(x + 3)$

4

$m = \frac{1}{2}, \quad P = (1, -5)$

4

$y + 5 = \frac{1}{2}(x - 1)$

5

$m = 9, \quad P = (-4, -4)$

5

$y + 4 = 9(x + 4)$

Use point–gradient form whenever you know a gradient and a point. No expansion is needed unless a particular form is requested.

Using 2 points

Solution 22

Example Find the line through (4,5) and (6,2), in standard form.

Example

$$m = \frac{2 - 5}{6 - 4} = -\frac{3}{2}$$

$$y - 5 = -\frac{3}{2}(x - 4)$$

$$3x + 2y - 22 = 0$$

Your turn Find the line through (−1,9) and (4,5), in standard form.

Your turn

$$m = -\frac{4}{5}$$

$$y - 9 = -\frac{4}{5}(x + 1)$$

$$4x + 5y - 41 = 0$$

Exercise 5C – Pure Y1 Page 94-95 Solution 23

3 Find the line through A(7,2), B(9,−8) in standard form.

3

$m = -5, \quad y - 2 = -5(x - 7)$

$5x + y - 37 = 0$

5 The line through (a,4), (3a,3) is x+6y+c=0. Find a,c.

5

$-\frac{1}{2a} = -\frac{1}{6} \Rightarrow a = 3$

$3 + 24 + c = 0 \Rightarrow c = -27$

6 The line through (7a,5), (3a,3) is x+by−12=0. Find a,b.

6

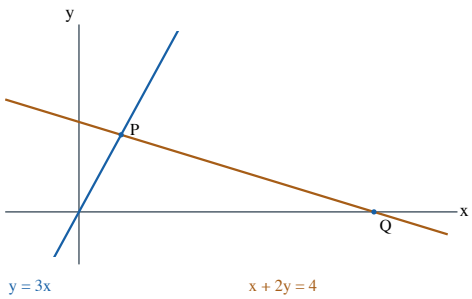
$y - 5 = \frac{1}{2a}(x - 7a)$

$x - 2ay + 3a = 0$

$3a = -12 \Rightarrow a = -4, \quad b = 8$

Intersection of lines Solution 24

The lines $y = 3x$ and $x + 2y = 4$ meet at P. The second meets the x-axis at Q.



a Find P.

a $x + 6x = 4 \Rightarrow P = (\frac{4}{7}, \frac{12}{7})$

b Find Q.

b

$y = 0 \Rightarrow Q = (4, 0)$

Test Your Understanding

Solution 25

a Find the equation of L_1 through $(-1,3)$, $(11,12)$. Use integer coefficients.

b Find its intersection with $L_2 : 3y + 4x - 30 = 0$.

a

$$m = \frac{9}{12} = \frac{3}{4}$$

$$y - 3 = \frac{3}{4}(x + 1)$$

$$4y - 3x - 15 = 0$$

b

$$12y - 9x = 45$$

$$12y + 16x = 120$$

$$25x = 75$$

$$(x, y) = (3, 6)$$

Exercise 5D – Pure Y1 Page 96

Solution 26

6 Find where the line through $(-3,-5)$ and $(4,9)$ meets the y-axis.

9 The lines y=4x−10 and y=x−1 meet at T. Find the line through T with gradient $-2/3$.

6

$$m = 2, \quad y + 5 = 2(x + 3)$$

$$y = 2x + 1 \Rightarrow G = (0, 1)$$

9

$$4x - 10 = x - 1 \Rightarrow T = (3, 2)$$

$$y - 2 = -\frac{2}{3}(x - 3)$$

$$2x + 3y - 12 = 0$$

Exercise 5D – Pure Y1 Page 96

Solution 27

14 Find the line through C(0,5) and D(−5,0).

14

$m = 1, \quad y = x + 5$

$x - y + 5 = 0$

16 The lines y=−2x+1 and y=x+7 meet at L. Find the line through L and M(−3,1).

16

$-2x + 1 = x + 7 \Rightarrow L = (-2, 5)$

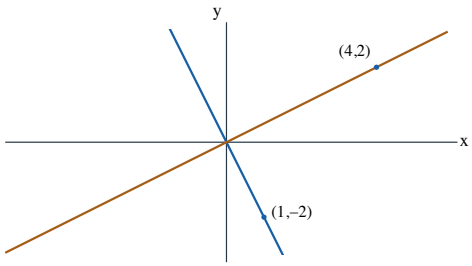
$m = \dfrac{5 - 1}{-2 + 3} = 4$

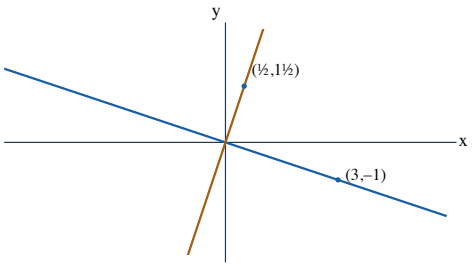
$y - 1 = 4(x + 3) \Rightarrow y = 4x + 13$

Gradients of perpendicular lines

Solution 28

Compare the gradients of these perpendicular pairs.





Pair 1 Calculate both gradients.

Pair 1

$-2, \quad \dfrac{1}{2}$

Pair 2 Calculate both gradients.

Pair 2

$-\dfrac{1}{3}, \quad 3$

If one rise/run is b/a, a 90° rotation gives −a/b. The gradients are negative reciprocals. This formula requires nonvertical, nonhorizontal lines.

Perpendicular Lines

Solution 29

Parallel lines have equal gradients. For perpendicular lines with defined gradients:

$m_1 m_2 = -1$

a

$m = 2$

a

$m_\perp = -\frac{1}{2}$

b

$m = -3$

b

$m_\perp = \frac{1}{3}$

c

$m = \frac{1}{4}$

c

$m_\perp = -4$

d

$m = 5$

d

$m_\perp = -\frac{1}{5}$

e

$m = -\frac{2}{7}$

e

$m_\perp = \frac{7}{2}$

f

$m = \frac{7}{5}$

f

$m_\perp = -\frac{5}{7}$

A horizontal line is perpendicular to a vertical line; handle this case directly.

Example Problems

Solution 30

1 Find the line through (9,10) perpendicular to y=3x+2.

1

$y - 10 = -\frac{1}{3}(x - 9)$

2 L_1 passes through A(1,3), B(3,−1). A perpendicular line through B meets the x-axis where?

2

$m_1 = -2, \quad y + 1 = \frac{1}{2}(x - 3)$

$y = 0 \Rightarrow (5, 0)$

3 Are $y = \frac{1}{2}x$ and 2x−y+4=0 parallel, perpendicular, or neither?

3

Gradients are 1/2 and 2. They are unequal and their product is 1, not −1. Therefore neither.

Test Your Understanding

Solution 31

1 A line through the origin is perpendicular to $y = -\frac{1}{2}x + 4$. Find their intersection A.

1

$y = 2x$

$2x = -\frac{1}{2}x + 4 \Rightarrow x = \frac{8}{5}$

$A = (\frac{8}{5}, \frac{16}{5})$

2 Find the line through (4,7) perpendicular to y=2x+2. Use standard form.

2

$y - 7 = -\frac{1}{2}(x - 4)$

$x + 2y - 18 = 0$

Exercise 5E/5F – Pure Y1 Page 96

Solution 32

3 A(−6,2), B(4,8), C(6,1), D(−9,−8) form a quadrilateral. Show it is a trapezium.

3

$m_{AB} = \frac{6}{10} = \frac{3}{5}, \quad m_{CD} = \frac{9}{15} = \frac{3}{5}$

AB is parallel to CD, so ABCD is a trapezium.

7 Find the line through the origin parallel to 2x−3y−1=0.

7

$m = \frac{2}{3}, \quad y = \frac{2}{3}x$

$2x - 3y = 0$

Exercise 5F – Pure Y1 Page 100

Solution 33

9 Show the lines through $(-3,0)$, $(3,-2)$ and through $(1,8)$, $(-1,2)$ are perpendicular.

9

$m_l = -\frac{1}{3}, \quad m_n = 3$

$m_l m_n = -1$

10 Show A$(-1,5)$, B$(7,1)$, C$(5,-3)$, D$(-3,1)$ form a rectangle.

10

$m_{AB} = m_{CD} = -\frac{1}{2}$

$m_{BC} = m_{DA} = 2$

Both pairs of opposite sides are parallel. Adjacent gradients multiply to -1, so the angles are right angles.

Exercise 5E/5F – Pure Y1 Page 96

Solution 34

1. MAT Reflect $(3,4)$ in 3x+4y=50.

1. MAT

$y - 4 = \frac{4}{3}(x - 3)$

The perpendicular meets the mirror at $(6,8)$, the midpoint of the point and its image.

$(2 \cdot 6 - 3, 2 \cdot 8 - 4) = (9, 12)$

2. MAT Reflect $(1,0)$ in y=mx.

2. MAT

The perpendicular foot is

$H = (\frac{1}{1+m^2}, \frac{m}{1+m^2})$

The image is 2H$-(1,0)$:

$(\frac{1-m^2}{1+m^2}, \frac{2m}{1+m^2})$

3. STEP For $A = (p_1, q_1), B = (p_2, q_2), C = (p_3, q_3)$, find where the medians from A and B meet and verify this point lies on the median from C. Let $H = (p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that AH perpendicular to BC implies $p_2^2 + q_2^2 = p_3^2 + q_3^2$. Give the analogous result for BH perpendicular to AC, then deduce CH perpendicular to AB.

3. STEP

$G = (\frac{p_1 + p_2 + p_3}{3}, \frac{q_1 + q_2 + q_3}{3})$

G is two-thirds along each median. Writing position vectors A,B,C:

$(B + C) \cdot (C - B) = |C|^2 - |B|^2 = 0$

Hence $p_2^2 + q_2^2 = p_3^2 + q_3^2$. Similarly BH perpendicular to AC gives $p_1^2 + q_1^2 = p_3^2 + q_3^2$. Thus

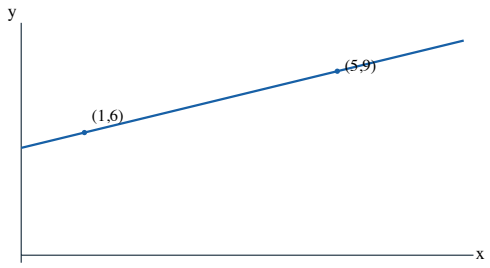
$(A + B) \cdot (B - A) = |B|^2 - |A|^2 = 0$

and CH is perpendicular to AB.

Distances between points

Solution 35

Find the distance between (1,6) and (5,9).



Distance

$\Delta x = 4, \quad \Delta y = 3$

$d = \sqrt{4^2 + 3^2} = 5$

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Use Pythagoras on the horizontal and vertical changes.

Examples

Solution 36

a Find the distance between (3,4) and (5,7).

a

$\sqrt{2^2 + 3^2} = \sqrt{13}$

b Find the distance between (5,1) and (6,−3).

b

$\sqrt{1^2 + 4^2} = \sqrt{17}$

c Find the distance between (0,−2) and (−1,3).

c

$\sqrt{1^2 + 5^2} = \sqrt{26}$

d Find the distance between (1,10) and (4,14).

d

$\sqrt{3^2 + 4^2} = 5$

e Find the distance between (3,−1) and (0,1).

e

$\sqrt{3^2 + 2^2} = \sqrt{13}$

f Find the distance between (−4,−2) and (−12,4).

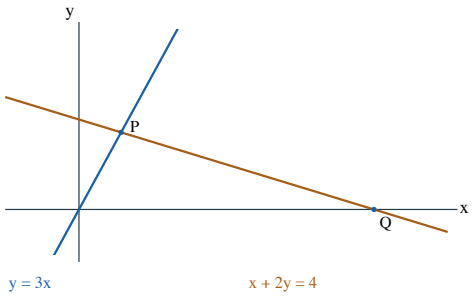
f

$\sqrt{8^2 + 6^2} = 10$

Area of Shapes

Solution 37

Lines y=3x and x+2y=4 meet at P. Q is the second line's x-intercept; O is the origin.



a Find P.

a

$$P = (\frac{4}{7}, \frac{12}{7})$$

b Find the area of triangle OPQ.

b

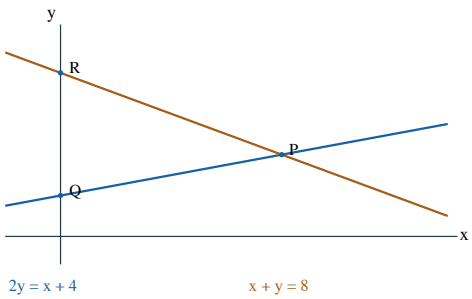
$$Q = (4, 0)$$

$$\text{Area} = \frac{1}{2}(4)(\frac{12}{7}) = \frac{24}{7}$$

Further Example

Solution 38

Lines 2y=x+4 and x+y=8 meet at P. Their y-intercepts are Q and R respectively.



a Find the length PQ.

a

$$P = (4, 4), \quad Q = (0, 2)$$

$$PQ = \sqrt{4^2 + 2^2} = \sqrt{20}$$

b Find the area PQR.

b

$$R = (0, 8), \quad QR = 6$$

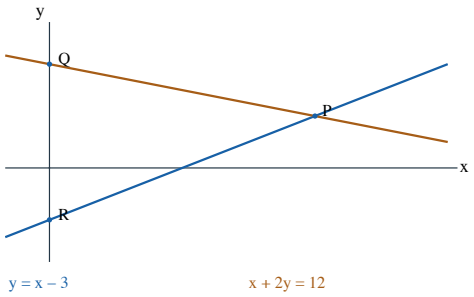
$$\text{Area} = \frac{1}{2}(6)(4) = 12$$

A vertical or horizontal side often makes a convenient base for a triangle.

Test Your Understanding

Solution 39

Lines y=x−3 and x+2y=12 meet at P. Q=(0,6), R=(0,−3) are their y-intercepts.



a Find P.

a $3x - 6 = 12 \Rightarrow P = (6, 3)$

b Find the area PQR.

b $\frac{1}{2}(9)(6) = 27$

c Find PQ.

c

$PQ = \sqrt{6^2 + 3^2} = 3\sqrt{5}$

Exercise 5G – Pure Y1 Page 102

Solution 40

7 P lies on y=4−3x and is $\sqrt{34}$ units from the origin. Find both possible positions.

7

$x^2 + (4 - 3x)^2 = 34$

$5x^2 - 12x - 9 = (5x + 3)(x - 3) = 0$

$P = (-\frac{3}{5}, \frac{29}{5})$ or $(3, -5)$

8 A(2,7), B(5,−6), C(8,−6). Find the side lengths, classify the triangle and find its area.

8

$AB = \sqrt{178},\ BC = 3,\ AC = \sqrt{205}$

All sides differ: scalene.

Area $= \frac{1}{2}(3)(13) = \frac{39}{2}$

Exercise 5G – Pure Y1 Page 103 Solution 41

a l_1 passes through $(-4,14)$ with gradient $-1/4$. Find its equation.

a

$$y - 14 = -\frac{1}{4}(x + 4)$$

$$x + 4y - 52 = 0$$

b $l_2 : y = 3x$ meets l_1 at B. Find B.

b

$$x + 12x = 52 \Rightarrow B = (4, 12)$$

c l_1 meets the y-axis at A. Find the area OAB.

c

$$A = (0, 13)$$

$$\text{Area} = \frac{1}{2}(13)(4) = 26$$

Exercise 5G – Pure Y1 Page 102-103 Solution 42

MAT Find the shortest distance from the origin to 3x+4y=25.

MAT

The perpendicular through the origin is y=4x/3.

$$3x + \frac{16}{3}x = 25 \Rightarrow (x, y) = (3, 4)$$

$$d = \sqrt{3^2 + 4^2} = 5$$

What's the point of straight line equations?

Solution 43

Linear relationships

Examples include a fixed salary plus commission, temperature conversions and simple altitude–temperature models.

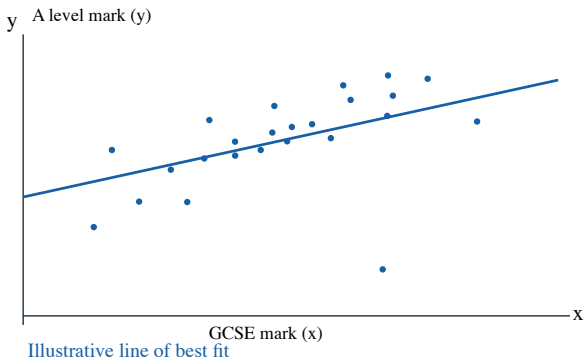
Arithmetic sequences

The sequence 2,5,8,11,14 has constant difference 3. Its nth term is $3n - 1$, a linear expression.

Look for a constant absolute change. Real data may only be approximately linear; specify the range where a model is reasonable.

Modelling

Solution 44



Illustrative line of best fit

A straight-line model

Plot paired GCSE and A level marks on a scatter diagram. A line of best fit can describe an approximate relationship.

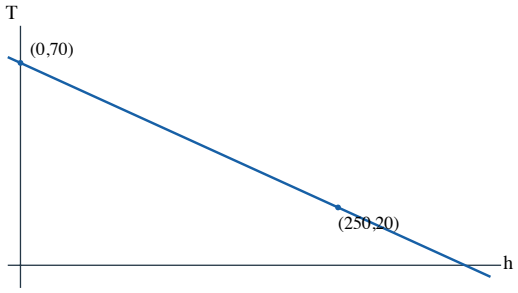
Interpret the parameters

The gradient estimates the change in predicted A level mark per extra GCSE mark. The intercept is the prediction when the GCSE mark is zero.

The intercept may have no practical meaning if zero lies far outside the data. Predictions are estimates; extrapolation beyond the observed range is less reliable.

Example

Solution 45



Temperature against altitude

a Temperature is 70˚F at altitude 0 m and 20˚F at altitude 250 m. Find a linear model T=mh+c.

a

$m = -0.2, \quad c = 70$

$T = 70 - 0.2h$

b Interpret m and c.

b

Temperature at sea level is 70˚F. Each extra metre reduces the predicted temperature by 0.2˚F.

c At what altitude does the model predict 0˚F?

c

$0 = 70 - 0.2h \Rightarrow h = 350$ m

This prediction extrapolates beyond the two given observations.

Evaluating if a linear model is most suitable

Solution 46

Population A population is 26,000 in 2017 and increases by 150 each year. Model it t years after 2017.

Population

$P = 26000 + 150t$

Assumption When is a linear model reasonable?

Assumption

It assumes a constant absolute annual increase. If growth is instead a constant percentage, an exponential model may be more appropriate.

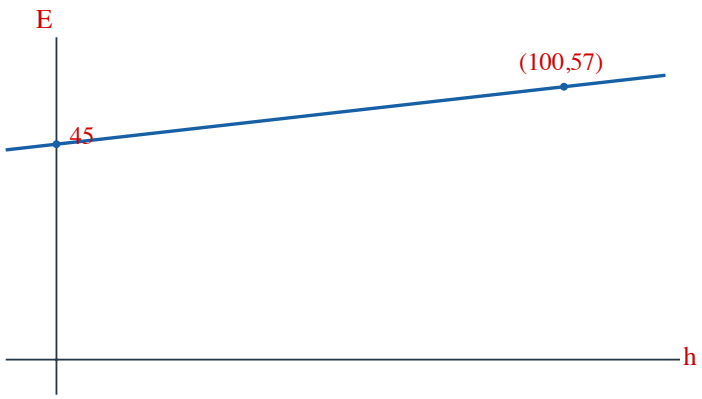
Exercise 5H – Pure Y1 Page 106-108 Solution 47

Electricity cost E (£) for usage h (kWh):

h	0	15	40	60	80	110
E	45	46.8	49.8	52.2	54.6	58.2

a. Draw a graph of the data.

Sketch



E = 0.12h + 45

b Why is a linear model appropriate?

b

The points lie on a straight line.

c Find an equation E=ah+b.

c

$E = 0.12h + 45$

d Interpret the coefficients.

d

12p per kWh plus a fixed charge of £45.

e Find the cost for 65 kWh.

e

$E = 0.12(65) + 45 = 52.80$

Cost: £52.80.

Exercise 5H – Pure Y1 Page 106-108 Solution 48

a 20°C corresponds to 68°F; 9°C to 48.2°F. Find F=mC+c.

a

$$m = \frac{68 - 48.2}{20 - 9} = 1.8$$

$F = 1.8C + 32$

b Convert a recorded temperature of 101.3°F to Celsius.

b

$$C = \frac{101.3 - 32}{1.8} = 38.5$$

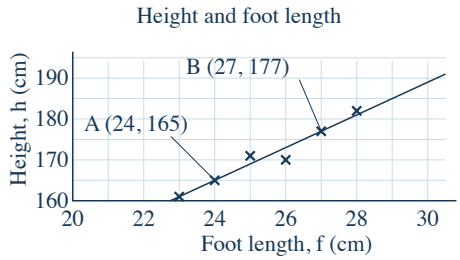
c At which temperature do the scales agree?

c

$C = 1.8C + 32 \Rightarrow C = -40$

$-40°\text{C} = -40°\text{F}.$

8. The scatter graph shows height h (cm) and foot length f (cm) for a group of students.



8(a) Why is a linear model suitable?

8(a) The points lie close to a straight line.

8(b) Use A(24,165), B(27,177) to find h=af+b.

8(b)

$$a = \frac{177 - 165}{27 - 24} = 4, \quad b = 69$$

$$h = 4f + 69$$

8(c) Predict h when f=26.5.

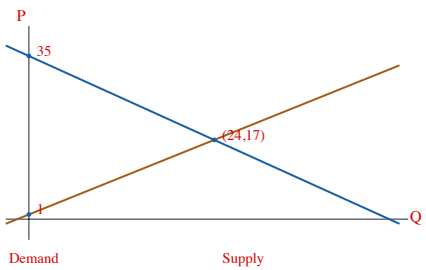
8(c)

$$h = 4(26.5) + 69 = 175 \text{ cm}$$

9. Demand: $P = -\frac{3}{4}Q + 35$. Supply: $P = \frac{2}{3}Q + 1$.

a. Sketch both lines on the same axes.

Sketch



9(b) Find the equilibrium price P and quantity Q.

9(b)

$$-\frac{3}{4}Q + 35 = \frac{2}{3}Q + 1$$

$$\frac{17}{12}Q = 34 \Rightarrow Q = 24, \; P = 17$$